RITAM MUKHERJEE

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EDUCATION

University of Waterloo Expected May 2030
Bachelor of Applied Science in Mechatronics Engineering *Waterloo, Ontario*
- **Relevant Coursework:** Data Structures and Algorithms (C++),Digital Logic, Computer Structures and OS, Linear Algebra

EXPERIENCE

Independent Research Project – Dr. Eugene Li Aug 2026 – Present
Undergraduate Student Researcher *Waterloo, Ontario*
- Building a multi-camera pose estimation pipeline fusing synchronized camera streams via **YOLOv8** and transformers; modeled **BSIP anthropometrics** and dynamic kinematics (CoM, ground reaction forces, 3D joint trajectories).

WATonomous Sep 2026 – Present
Core Member - Humanoid Perception *Waterloo, Ontario*
- Building real-time perception and teleoperation data pipelines in **C++** and **ROS 2** for long-horizon humanoid manipulation tasks.
- Designing multi-modal sensory ingestion workflows to synchronize visual/joint telemetry and streamline policy demonstration recording.

SuraNeoAge Systems Transformations (Early Stage Startup) May 2026 – Aug 2026
Software Engineering Intern *Remote*
- Fine-tuned Phi-4 Mini (3.8B params) via **QLoRA** for financial entity and topic extraction, using GPT-4o-generated ground-truth labels across 2,700+ scraped financial news articles enriched with National Stock Exchange sector data; outperformed the base model and GPT-4o mini in a 100-article blind ABC comparison judged by the team's business analyst.
- Built the ingestion-to-training data pipeline (PostgreSQL, FastAPI, Docker) and fine-tuned FinBERT (110M params) for company NER, reaching **0.76 token F1 / 0.88 recall**.
- Built an on-demand LLM relevance-scoring endpoint replacing manual editorial curation; persisted latest verdict per article in **Redis** with automatic cache invalidation on reclassification, decoupling read latency from LLM inference time

University of Waterloo – NanoTech lab and RoboHub Sep 2025 – May 2026
Undergraduate Student Researcher *Waterloo, Ontario*
- Worked under Dr. Shirley Tang on early-stage development of a 7-axis 3D printer using a Gen3 robot arm — implemented **ROS 2 Cartesian trajectory-tracking** and homing scripts on the physical arm, while researching non-planar slicing and g-code conversion

University of Waterloo – WATiMAKE lab Aug 2024 – Jun 2025
Volunteer Student Researcher *Waterloo, Ontario*
- Assisted Dr. Eugene Li on an industrial CNC tool remaining-useful-life project — built a sensor-to-model data pipeline, prototyped a PyTorch Transformer, and trained an **NCA+KNN classifier** reaching **79.6%** test accuracy on 1.5M+ accelerometer samples

PROJECTS

Personal Website | *https://www.ritammukherjee.ca/ — Next.js, React, TypeScript, Supabase, Vercel*
- Developed a full-stack portfolio featuring live integrations with Spotify and Discord APIs to display real-time user activity
- Implemented a custom messaging system using Supabase for the backend and integrated GitHub data to showcase dynamic project updates and repository statistics.

Axiom | *https://github.com/forkiron/axiom — Node.js, TypeScript, Supabase, Vercel*
- Built an AI platform to standardize Canadian school grades via test-difficulty analysis, a 3D interactive map, and a natural-language agent
- Developed full-stack with Next.js, Gemini AI, Mapbox, and Supabase with crowd-sourced adjustment data persistence

Motion Activated Door Monitoring System | *https://github.com/ryanli222/smartdoorbell — Raspberry Pi, OpenCV, Docker, Ngrok, FastAPI*
- Developed a real-time motion detection system using OpenCV on a Raspberry Pi to monitor a webcam feed and detect door activity.
- On detection, activated a display, streamed live video, played alert audio, enabled two-way audio, and uploaded images to FastAPI via Ngrok

1000 Elo Chessbot | *https://github.com/j3rry1iu/ChessHacks-Training — Python, CNNs, Numpy, HuggingFace*
- Built a chess engine using a ResNet style CNN, policy/value head, and a depth3 search to generate moves within 250ms.
- Used Modal cloud computing to train the model, uploading model weights and storing the model on HuggingFace.

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript, JavaScript, SQL
ML/Data: PyTorch, HuggingFace, QLoRA, scikit-learn, Numpy, OpenCV, TensorFlow, Reinforcement Learning
Robotics/Controls: ROS 2, Foxglove Studio, Kinematics (FK/IK), Trajectory Planning, State Estimation (EKF/UKF), SolidWorks
Web/Infra: FastAPI, Next.js, React, PostgreSQL, Redis, Supabase, Docker, Vercel, Git, Bash, Linux